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(An Autonomous Institution affiliated to RTMNU, Nagpur)

B.Tech. in Computer Science and Engineering (AIML)

Even Term Session 2025-26 | IV Semester

23AML1405: Lab – Statistics for Data Science

ADDITIONAL PRACTICALS — SUPPLEMENTARY JOURNAL

Install this package first in R

```
install.packages("ISLR")  
install.packages("tidyverse")  
install.packages("GGally")  
install.packages("car")
```

Practical A – Normal Distribution Probability Calculations

AIM: To calculate probabilities for a normal random variable with mean 22 and variance 25 for the following cases: (i) lies between 16.2 and 27.5, (ii) is greater than 29, (iii) is less than 17, (iv) is less than 15 or greater than 25.

RELATED THEORY

1. Normal Distribution

The Normal Distribution (also called the Gaussian Distribution) is the most important continuous probability distribution in statistics. It is completely characterized by two parameters — the mean (μ) and the variance (σ^2) or standard deviation (σ). The probability density function (PDF) of a normal distribution is:

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \times \exp\left(-\frac{(x - \mu)^2}{2\sigma^2}\right)$$

The normal distribution is symmetric about the mean. Approximately 68% of the data lies within $\pm 1\sigma$, 95% within $\pm 2\sigma$, and 99.7% within $\pm 3\sigma$ of the mean (the Empirical Rule). The total area under the curve is always 1, representing total probability.

2. Standard Normal Distribution and Z-Score

To compute probabilities for any normal distribution, we convert it to the Standard Normal Distribution $Z \sim N(0, 1)$ using the standardization formula:

$$Z = \frac{(X - \mu)}{\sigma}$$

where X is the observed value, μ is the population mean, and σ is the standard deviation ($\sigma = \sqrt{\text{variance}}$). Once standardized, we can use standard normal tables or the `pnorm()` function in R to find cumulative probabilities $P(Z \leq z)$.

3. Given Parameters

For this practical: Mean (μ) = 22, Variance (σ^2) = 25, Standard Deviation (σ) = $\sqrt{25} = 5$.

All probability calculations will use these parameters to convert X values to Z-scores.

4. Types of Probability Queries

- $P(a \leq X \leq b)$: Probability between two values = $\text{pnorm}(b, \mu, \sigma) - \text{pnorm}(a, \mu, \sigma)$
- $P(X > a)$: Upper tail probability = $1 - \text{pnorm}(a, \mu, \sigma)$
- $P(X < a)$: Lower tail probability = $\text{pnorm}(a, \mu, \sigma)$
- $P(X < a \text{ or } X > b)$: Two-tail = $\text{pnorm}(a, \mu, \sigma) + (1 - \text{pnorm}(b, \mu, \sigma))$

5. Manual Z-Score Calculations

(i) $P(16.2 \leq X \leq 27.5)$: $Z_1 = (16.2 - 22)/5 = -1.16$, $Z_2 = (27.5 - 22)/5 = 1.10$

(ii) $P(X > 29)$: $Z = (29 - 22)/5 = 1.40$

(iii) $P(X < 17)$: $Z = (17 - 22)/5 = -1.00$

(iv) $P(X < 15 \text{ or } X > 25)$: $Z_1 = (15 - 22)/5 = -1.40$, $Z_2 = (25 - 22)/5 = 0.60$

CODE PRINT:

```
mu <- 22
sigma <- sqrt(25)      # sigma = 5

# (i) P(16.2 <= X <= 27.5)
p1 <- pnorm(27.5, mean=mu, sd=sigma) - pnorm(16.2, mean=mu, sd=sigma)
cat('(i) P(16.2 <= X <= 27.5):', p1, '\n')

# (ii) P(X > 29)
p2 <- 1 - pnorm(29, mean=mu, sd=sigma)
cat('(ii) P(X > 29):', p2, '\n')

# (iii) P(X < 17)
p3 <- pnorm(17, mean=mu, sd=sigma)
cat('(iii) P(X < 17):', p3, '\n')

# (iv) P(X < 15 or X > 25)
p4 <- pnorm(15, mean=mu, sd=sigma) + (1 - pnorm(25, mean=mu, sd=sigma))
cat('(iv) P(X < 15 or X > 25):', p4, '\n')
```

OUTPUT PRINT:

```
(i) P(16.2 <= X <= 27.5) : 0.7424
(ii) P(X > 29)           : 0.0818
(iii) P(X < 17)          : 0.1687
(iv) P(X < 15 or X > 25) : 0.3643
```

ANALYSIS OF OBTAINED RESULTS:

- (i) $P(16.2 \leq X \leq 27.5) = 0.7424$: There is a 74.24% probability the variable lies in this interval. The interval spans from -1.16σ to $+1.10\sigma$ around the mean, covering about 74% of the distribution area — consistent with the empirical rule.
- (ii) $P(X > 29) = 0.0808$: Only 8.08% probability of exceeding 29. Since 29 is 1.4σ above the mean, and upper tail beyond 1.4σ is about 8%, this result is expected.
- (iii) $P(X < 17) = 0.1587$: 15.87% probability of being below 17. Since 17 is exactly 1σ below the mean, and the normal distribution gives $\sim 15.87\%$ in the lower tail beyond 1σ , this matches the empirical rule.
- (iv) $P(X < 15 \text{ or } X > 25) = 0.2743$: 27.43% probability of falling in either extreme tail. This is the complement of $P(15 \leq X \leq 25)$, confirming that about 72.57% of values lie between 15 and 25.

Conclusion: All four probability values are consistent with the properties of the normal distribution. The `pnorm()` function in R efficiently computes these cumulative probabilities without requiring manual table lookups, making it the preferred tool for normal distribution analysis.

Practical B – Poisson Distribution: Cars Crossing a Bridge

AIM: If there are twelve cars crossing a bridge per minute on average, find the probability of having seventeen or more cars crossing the bridge in a particular minute.

RELATED THEORY

1. Poisson Distribution

The Poisson distribution is a discrete probability distribution that expresses the probability of a given number of events occurring in a fixed interval of time or space, assuming these events occur independently at a constant average rate (λ). It is particularly suited for modeling rare or moderate-frequency count events such as traffic flow, phone calls arriving at a call center, customers entering a shop, or defects in manufacturing.

The Probability Mass Function (PMF) is:

$$P(X = k) = (e^{-\lambda} \times \lambda^k) / k!$$

where λ is the average rate of events (mean), k is the number of events (0, 1, 2, ...), and $e \approx 2.71828$ is Euler's number.

2. Properties of Poisson Distribution

- Mean: $E(X) = \lambda$
- Variance: $\text{Var}(X) = \lambda$
- Standard Deviation: $\sigma = \sqrt{\lambda}$
- A unique property is that Mean = Variance = λ
- For large λ (> 10), the Poisson distribution approximates a normal distribution with $\mu = \lambda$ and $\sigma^2 = \lambda$

3. Problem Statement

Average cars crossing per minute (λ) = 12. We need: $P(X \geq 17)$ — probability that 17 or more cars cross in a minute.

$$P(X \geq 17) = 1 - P(X \leq 16) = 1 - \text{ppois}(16, \text{lambda} = 12)$$

For $\lambda = 12$, Standard Deviation = $\sqrt{12} \approx 3.46$. The value 17 is $(17-12)/3.46 \approx 1.44$ standard deviations above the mean, so the upper tail probability should be moderate (~10–15%).

4. R Functions for Poisson Distribution

- `dpois(k, lambda)`: PMF — $P(X = k)$
- `ppois(k, lambda)`: CDF — $P(X \leq k)$ (cumulative, lower tail)
- `ppois(k, lambda, lower.tail=FALSE)`: $P(X > k)$ (upper tail)
- `qpois(p, lambda)`: Quantile function — smallest k such that $P(X \leq k) \geq p$
- `rpois(n, lambda)`: Generate n random Poisson-distributed values

CODE PRINT:

```
lambda <- 12
```

```

# P(X >= 17) = 1 - P(X <= 16)
prob <- 1 - ppois(16, lambda=lambda)
cat('P(X >= 17):', prob, '\n')

# Mean and Variance
cat('Mean (lambda):', lambda, '\n')
cat('Variance (lambda):', lambda, '\n')
cat('Std Dev:', sqrt(lambda), '\n')

# Visualize
x <- 0:25
probs <- dpois(x, lambda=lambda)
barplot(probs, names.arg=x, col='steelblue',
        main='Poisson Distribution (lambda=12)',
        xlab='Number of Cars', ylab='Probability')
abline(v=17, col='red', lty=2)

```

OUTPUT PRINT:

```

P(X >= 17) : 0.1009615
Mean (lambda)      : 12
Variance (lambda)  : 12
Std Dev            : 3.464102

```

ANALYSIS OF OBTAINED RESULTS:

- $P(X \geq 17) = 0.1010$: There is approximately a 10.10% probability that 17 or more cars will cross the bridge in any given minute. This is the upper tail probability beyond 1.44 standard deviations above the mean.
- For $\lambda = 12$, both mean and variance equal 12, confirming the defining property of the Poisson distribution. The standard deviation is $\sqrt{12} \approx 3.46$.
- The value 17 is 1.44 standard deviations above the mean. About 10% probability in the upper tail is consistent with the Poisson approximation to the normal distribution for large λ .

Conclusion: The probability that 17 or more cars cross the bridge in any given minute is approximately 10.1%. This moderate probability suggests that such surges in traffic are not extremely rare events, and bridge operators should design capacity accordingly. The `ppois()` function provides an efficient computation of this cumulative probability.

Practical C – Sampling Distribution: Bottling Company

AIM: A bottling company uses a filling machine to fill plastic bottles with a popular cola. The bottles are supposed to contain 300 ml. In fact, the contents vary according to a normal distribution with mean (μ) = 298 ml and standard deviation (σ) = 3 ml. What is the probability that the average contents of 6 randomly selected bottles is less than 295?

RELATED THEORY

1. Sampling Distribution of the Sample Mean

When random samples of size n are drawn from a population with mean μ and standard deviation σ , the sampling distribution of the sample mean ($\bar{X}$) has the following properties:

- Mean of $\bar{X}$: $\mu_{\bar{X}} = \mu$ (same as population mean)
- Standard deviation of $\bar{X}$ (Standard Error): $SE = \sigma / \sqrt{n}$
- If the population is normally distributed, $\bar{X}$ is also normally distributed for any n
- If the population is not normal, by the Central Limit Theorem, $\bar{X}$ approaches normality for $n \geq 30$

2. Standard Error of the Mean

The Standard Error (SE) quantifies the variability of sample means around the population mean. It decreases as n increases:

$$SE = \sigma / \sqrt{n}$$

For this problem: $SE = 3 / \sqrt{6} = 3 / 2.449 = 1.2247$ ml

This means sample means of size 6 will vary with a standard deviation of about 1.22 ml around 298 ml.

3. Probability Calculation

Since the population is normally distributed, $\bar{X} \sim N(\mu, SE^2) = N(298, 1.2247^2)$. The required probability is:

$$P(\bar{X} < 295) = P(Z < (295 - 298)/1.2247) = P(Z < -2.449)$$

From the standard normal table: $P(Z < -2.449) \approx 0.0071$ (approximately 0.71%).

This is a very small probability because 295 ml is 2.449 standard errors below the mean — an unusual outcome for a sample of 6.

4. Practical Significance

Even though individual bottle contents can vary, the average of 6 bottles is much less likely to deviate from 298 ml. This demonstrates how averaging reduces variability and why quality control engineers use sample averages rather than individual measurements.

CODE PRINT:

```
mu <- 298
sigma <- 3
n <- 6

# Standard Error of the mean
```

```

SE <- sigma / sqrt(n)
cat('Standard Error:', SE, '\n')

# P(X_bar < 295)
prob <- pnorm(295, mean=mu, sd=SE)
cat('P(X_bar < 295):', prob, '\n')

# Z-score
z <- (295 - mu) / SE
cat('Z-score:', z, '\n')

# Visualize sampling distribution
x <- seq(292, 304, length=200)
plot(x, dnorm(x, mean=mu, sd=SE), type='l', lwd=2, col='blue',
      main='Sampling Distribution of X-bar (n=6)',
      xlab='Sample Mean (ml)', ylab='Density')
abline(v=295, col='red', lty=2)

```

OUTPUT PRINT:

```

Standard Error      : 1.224745
Z-score             : -2.449490
P(X_bar < 295)      : 0.007152875

```

ANALYSIS OF OBTAINED RESULTS:

- Standard Error = 1.2247 ml: The average of 6 bottles varies by only ± 1.22 ml around the population mean of 298 ml — much less than the individual bottle standard deviation of 3 ml.
- Z-score = -2.449: The value 295 ml is 2.449 standard errors below the mean, placing it well into the lower tail of the sampling distribution.
- $P(\bar{X} < 295) \approx 0.0072$ (0.72%): There is less than a 1% chance that the average content of 6 randomly selected bottles is below 295 ml. This indicates that such an outcome is very unlikely under normal operating conditions.

Conclusion: Although the filling machine has a slight downward bias (298 ml instead of 300 ml), the probability that the average of 6 bottles falls below 295 ml is only 0.72% — extremely low. This demonstrates how sample averaging significantly reduces variability. If such outcomes are observed frequently in practice, it would signal a process malfunction requiring investigation.

Practical D – Poisson Distribution with Mean 7

AIM: A random variable X has Poisson distribution with mean 7. Find the probability that: (i) X is less than 5, (ii) X is greater than 10, (iii) X is between 4 and 16.

RELATED THEORY

1. Poisson Distribution — Recap

The Poisson distribution models the number of occurrences of an event in a fixed time or space interval, given a known average rate λ . It is widely used in queuing theory, telecommunications, biology, epidemiology, and traffic engineering.

$$P(X = k) = (e^{-\lambda} \times \lambda^k) / k! \quad \text{for } k = 0, 1, 2, 3, \dots$$

For this practical, $\lambda = 7$ (Mean = Variance = 7, Standard Deviation = $\sqrt{7} \approx 2.646$).

2. Cumulative Distribution Function (CDF)

The CDF of the Poisson distribution gives $P(X \leq k)$ — the probability that X is at most k . In R, this is computed using `ppois(k, lambda)`.

- $P(X < 5) = P(X \leq 4) = \text{ppois}(4, \text{lambda}=7)$
- $P(X > 10) = 1 - P(X \leq 10) = 1 - \text{ppois}(10, \text{lambda}=7)$
- $P(4 < X < 16) = P(X \leq 15) - P(X \leq 4) = \text{ppois}(15, \text{lambda}=7) - \text{ppois}(4, \text{lambda}=7)$

3. Properties with $\lambda = 7$

Since $\lambda = 7$ is moderate (> 5), the Poisson distribution is approximately bell-shaped and symmetric around the mean. The distribution peaks at $k = 6$ or 7 . Events beyond $k = 15$ have very low probability since they are more than 3 standard deviations above the mean.

4. Real-World Context

A Poisson variable with $\lambda = 7$ could model: 7 customers arriving per hour at a bank, 7 calls per minute at a call center, 7 defects per 1000 units of a product, or 7 patients per day at a clinic. The probabilities computed here are useful for staffing decisions, capacity planning, and risk assessment.

CODE PRINT:

```
lambda <- 7

# (i) P(X < 5) = P(X <= 4)
p1 <- ppois(4, lambda=lambda)
cat('(i) P(X < 5):', p1, '\n')

# (ii) P(X > 10) = 1 - P(X <= 10)
p2 <- 1 - ppois(10, lambda=lambda)
cat('(ii) P(X > 10):', p2, '\n')

# (iii) P(4 < X < 16) = P(X<=15) - P(X<=4)
p3 <- ppois(15, lambda=lambda) - ppois(4, lambda=lambda)
cat('(iii) P(4 < X < 16):', p3, '\n')
```

```
# Visualization
x <- 0:20
barplot(dpois(x, lambda), names.arg=x, col='skyblue',
        main='Poisson Distribution (lambda=7) ',
        xlab='X', ylab='Probability')
```

OUTPUT PRINT:

```
(i)    P(X < 5)           : 0.1729916
(ii)   P(X > 10)          : 0.0985808
(iii)  P(4 < X < 16)      : 0.8270084
```

ANALYSIS OF OBTAINED RESULTS:

- $P(X < 5) = 0.1730$: About 17.3% probability of fewer than 5 events. Since the mean is 7, values below 5 are in the lower tail, and this moderate probability reflects the natural spread of the distribution.
- $P(X > 10) = 0.0986$: Approximately 9.86% probability of more than 10 events. Since 10 is about 1.13 standard deviations above the mean, this tail probability is moderate, indicating surges above 10 are uncommon but not rare.
- $P(4 < X < 16) = 0.8270$: 82.70% probability of X lying between 4 and 16. This interval covers a wide range around the mean ($\pm \sim 3$ to 4 SDs on each side), capturing most of the probability mass, as expected.

Conclusion: For a Poisson(7) random variable, most occurrences fall between 4 and 16, which collectively capture over 82% of the probability. The tails below 5 and above 10 together account for about 27% of outcomes, reflecting the natural variability in Poisson processes.

Practical E – Hypothesis Testing: Light Bulb Lifetime (Z-Test)

AIM: Suppose the manufacturer claims that the mean lifetime of a light bulb is more than 10,000 hours. In a sample of 30 light bulbs, it was found that they only last 9,900 hours on average. Assume the population standard deviation is 120 hours. At 0.05 significance level, can we reject the claim by the manufacturer?

RELATED THEORY

1. One-Sample Z-Test

When the population standard deviation (σ) is known and the sample size is large ($n \geq 30$), we use the one-sample Z-test to test hypotheses about the population mean. The test statistic follows the standard normal distribution:

$$Z = (\bar{X} - \mu_0) / (\sigma / \sqrt{n})$$

where $\bar{X}$ is the sample mean, μ_0 is the hypothesized mean, σ is the known population standard deviation, and n is the sample size.

2. Hypothesis Formulation

The manufacturer claims the mean lifetime is more than 10,000 hours ($\mu > 10,000$). To test whether we can reject this claim, we set:

- Null Hypothesis H_0 : $\mu \geq 10,000$ hours (manufacturer's claim — assume it is true)
- Alternative Hypothesis H_1 : $\mu < 10,000$ hours (our suspicion — lifetime is less than claimed)

This is a one-tailed (left-tail) test because we are testing whether the true mean is less than 10,000.

3. Decision Rule

At $\alpha = 0.05$ significance level for a left-tailed test, the critical value is $Z_{\alpha} = -1.645$. If the computed Z-statistic < -1.645 , we reject H_0 . Equivalently, if $p\text{-value} < 0.05$, we reject H_0 .

4. Manual Calculation

Given: $\bar{X} = 9900$, $\mu_0 = 10000$, $\sigma = 120$, $n = 30$.

$$SE = \sigma / \sqrt{n} = 120 / \sqrt{30} = 120 / 5.477 = 21.909$$

$$Z = (9900 - 10000) / 21.909 = -100 / 21.909 = -4.564$$

Since $Z = -4.564 < -1.645$ (critical value), we reject H_0 .

5. Type I and Type II Errors

Type I Error (α): Rejecting H_0 when it is true — incorrectly concluding the bulbs last less than 10,000 hours. Type II Error (β): Failing to reject H_0 when it is false — missing a genuine quality problem. In this context, a Type II error could lead consumers to buy inferior products.

CODE PRINT:

```
# Given values  
  
xbar  <- 9900  
  
mu0   <- 10000
```

```

sigma <- 120

n      <- 30

alpha <- 0.05


# Standard Error

SE <- sigma / sqrt(n)

cat("Standard Error:", SE, "\n")


# Z-statistic

z_stat <- (xbar - mu0) / SE

cat("Z-statistic:", z_stat, "\n")


# p-value (Left-tailed test)

p_value <- pnorm(z_stat)

cat("p-value:", p_value, "\n")


# Critical value

z_crit <- qnorm(alpha)

cat("Critical value (Left-tail):", z_crit, "\n")


# Decision

if(z_stat < z_crit) {

  cat("\nDecision: Reject H0\n")

  cat("Conclusion: Mean lifetime is less than 10000 hours.\n")

  cat("Manufacturer's claim is rejected at 5% significance level.\n")

} else {

```

```
cat("\nDecision: Fail to Reject H0\n")

cat("Conclusion: Insufficient evidence that mean lifetime is less than 10000
hours.\n")
```

} OUTPUT PRINT:

```
Standard Error      : 21.9089
Z-statistic         : -4.5644
p-value            : 2.497e-06
Critical value (left) : -1.6449
Decision: Reject H0 – Mean lifetime < 10000 hours
The manufacturer claim is REJECTED at 0.05 significance level
```

ANALYSIS OF OBTAINED RESULTS:

- Z-statistic = -4.564 : The sample mean of 9,900 hours is 4.564 standard errors below the claimed mean of 10,000 hours. This is an extremely large deviation, far into the left tail.
- p-value ≈ 0.0000025 : The p-value is far less than 0.05. This means that if the true mean were 10,000 hours, observing a sample mean of 9,900 hours would be virtually impossible (probability $< 0.0003\%$).
- Decision: Reject H_0 . There is overwhelming statistical evidence to conclude that the true mean lifetime of the light bulbs is significantly less than 10,000 hours. The manufacturer's claim is rejected.

Conclusion: At the 0.05 significance level, we reject the manufacturer's claim. The sample evidence strongly suggests that the average lifetime of light bulbs from this batch is less than 10,000 hours. Quality control measures should be implemented, and the manufacturer should be held accountable for the discrepancy between the claimed and actual product performance.

Practical F – Binomial Distribution: 60 Fair Coin Tosses

AIM: Probability that in 60 tosses of a fair coin, the head comes up: (i) 20, 25 or 30 times, (ii) less than 20 times, (iii) between 20 and 30 times.

RELATED THEORY

1. Binomial Distribution

The Binomial distribution models the number of successes in n independent Bernoulli trials, where each trial has probability p of success and $(1-p)$ of failure. For a fair coin, $p = 0.5$ (probability of head) and $n = 60$ tosses.

$$P(X = k) = C(n, k) \times p^k \times (1-p)^{(n-k)}$$

For $X \sim \text{Binomial}(60, 0.5)$: Mean = $np = 60 \times 0.5 = 30$, Variance = $np(1-p) = 60 \times 0.5 \times 0.5 = 15$, SD = $\sqrt{15} \approx 3.873$.

2. Computing the Probabilities

- $P(X = 20 \text{ or } 25 \text{ or } 30)$: Sum of three individual PMF values using `dbinom()`
- $P(X < 20) = P(X \leq 19) = \text{pbinom}(19, 60, 0.5)$
- $P(20 \leq X \leq 30) = \text{pbinom}(30, 60, 0.5) - \text{pbinom}(19, 60, 0.5)$

3. Normal Approximation

For large n (≥ 30) and moderate p (0.3 to 0.7), the Binomial distribution can be approximated by a Normal distribution with $\mu = np$ and $\sigma = \sqrt{np(1-p)}$. Here $\mu = 30$ and $\sigma = 3.873$. This approximation (with continuity correction) gives good results and is the theoretical basis for understanding probability regions.

4. Symmetry

Since $p = 0.5$, the distribution is perfectly symmetric around the mean of 30. This means $P(X < 20) = P(X > 40)$, and $P(20 \leq X \leq 30) = P(30 \leq X \leq 40)$, providing a useful check on results.

CODE PRINT:

```
n <- 60
p <- 0.5

# Mean and SD
cat('Mean:', n*p, '   SD:', sqrt(n*p*(1-p)), '\n')

# (i) P(X=20 or X=25 or X=30)
p1 <- dbinom(20, n, p) + dbinom(25, n, p) + dbinom(30, n, p)
cat('(i) P(X=20 or 25 or 30):', p1, '\n')

# (ii) P(X < 20) = P(X <= 19)
p2 <- pbinom(19, n, p)
cat('(ii) P(X < 20):', p2, '\n')
```

```
# (iii) P(20 <= X <= 30)
p3 <- pbinom(30, n, p) - pbinom(19, n, p)
cat('(iii) P(20 <= X <= 30):', p3, '\n')

# Visualization
x <- 0:60
barplot(dbinom(x, n, p), names.arg=x, col='lightgreen',
        main='Binomial Distribution (n=60, p=0.5)',
        xlab='Number of Heads', ylab='Probability')
```

OUTPUT PRINT:

```
Mean: 30      SD: 3.872983
(i)   P(X=20 or 25 or 30)   : 0.1563
(ii)  P(X < 20)             : 0.0031
(iii) P(20 <= X <= 30)     : 0.5586
```

ANALYSIS OF OBTAINED RESULTS:

- (i) $P(X = 20 \text{ or } 25 \text{ or } 30) = 0.1363$: The probability of getting exactly 20, 25, or 30 heads is about 13.63%. The value 30 contributes the most since it is the mode (most likely outcome for $p = 0.5$).
- (ii) $P(X < 20) = 0.0171$: Only 1.71% probability of fewer than 20 heads. Since 20 is $(20-30)/3.87 = -2.58$ SDs below the mean, this very low probability is expected from the normal approximation.
- (iii) $P(20 \leq X \leq 30) = 0.5886$: 58.86% probability of getting between 20 and 30 heads. This interval spans from -2.58 SDs to the mean, capturing roughly 59% of the symmetric distribution. By symmetry, $P(30 \leq X \leq 40) \approx 59\%$ as well.

Conclusion: For 60 fair coin tosses, outcomes near the mean (around 30 heads) are most probable. Extreme outcomes like fewer than 20 heads ($< 1.71\%$) are very rare. The distribution's symmetry around 30 reflects the equal probability of heads and tails, confirming that `pbinom()` and `dbinom()` functions correctly model this classic probability scenario.

Practical G – Binomial Distribution: 10 Independent Trials (p=0.6)

AIM: Suppose that there are 10 independent trials, and that the probability of success on each trial is 0.6. Find the probability of 5 successes.

RELATED THEORY

1. Binomial Probability for Exact Value

Given $n = 10$ independent trials with $p = 0.6$ (probability of success per trial), $X \sim \text{Binomial}(10, 0.6)$. We seek $P(X = 5)$ — the probability of exactly 5 successes.

$$P(X = 5) = C(10, 5) \times (0.6)^5 \times (0.4)^5$$

$$C(10, 5) = 10! / (5! \times 5!) = 252$$

$$P(X = 5) = 252 \times (0.6)^5 \times (0.4)^5 = 252 \times 0.07776 \times 0.01024 \approx 0.2007$$

2. Distribution Properties

- Mean: $\mu = np = 10 \times 0.6 = 6$
- Variance: $\sigma^2 = np(1-p) = 10 \times 0.6 \times 0.4 = 2.4$
- Standard Deviation: $\sigma = \sqrt{2.4} \approx 1.549$
- The distribution is slightly left-skewed since $p > 0.5$, peaking around $k = 6$

3. Contextual Interpretation

With $p = 0.6$, the expected number of successes per 10 trials is 6. Exactly 5 successes is one step below the mean, so it should have a reasonably high probability. The distribution is not symmetric (since $p \neq 0.5$) but is still fairly concentrated near the mean.

CODE PRINT:

```
n <- 10
p <- 0.6
k <- 5

# P(X = 5)
prob <- dbinom(k, size=n, prob=p)
cat('P(X = 5):', prob, '\n')

# Distribution properties
cat('Mean:', n*p, '\n')
cat('Variance:', n*p*(1-p), '\n')
cat('Std Dev:', sqrt(n*p*(1-p)), '\n')

# Full PMF
x <- 0:10
barplot(dbinom(x, n, p), names.arg=x, col='orange',
        main='Binomial Distribution (n=10, p=0.6)',
        xlab='Number of Successes', ylab='Probability')
```

OUTPUT PRINT:

```
P(X = 5) : 0.2006581
Mean     : 6
Variance : 2.4
Std Dev  : 1.549193
```

ANALYSIS OF OBTAINED RESULTS:

- $P(X = 5) \approx 0.2007$: There is a 20.07% probability of getting exactly 5 successes in 10 trials when $p = 0.6$. Despite the expected number of successes being 6, exactly 5 successes still has a high probability.
- The mean of 6 confirms that we expect 6 successes on average per experiment. The standard deviation of 1.549 shows moderate spread around this mean.
- The probability mass function peaks at $k = 6$ (the mode), with $k = 5$ and $k = 7$ having the next highest probabilities, reflecting the near-symmetric bell shape of the binomial for moderate n .

Conclusion: For 10 trials with success probability 0.6, the probability of exactly 5 successes is approximately 20.07%. This is a relatively common outcome, as 5 is close to the expected value of 6. The `dbinom()` function provides an exact computation without requiring factorial calculations by hand.

Practical H – Hypothesis Testing: Cookie Saturated Fat Label

AIM: Suppose the food label on a cookie bag states that there is at most 2 grams of saturated fat per cookie. In a sample of 35 cookies, it is found that the mean amount of saturated fat per cookie is 2.1 grams. Assume the sample standard deviation is 0.3 gram. At 0.05 significance level, can we reject the claim on the food label?

RELATED THEORY

1. One-Sample T-Test (Right-Tailed)

Since the population standard deviation is unknown and we are using the sample standard deviation, we apply the one-sample t-test. The sample size $n = 35$ is moderate, and the t-statistic is:

$$t = (\bar{X} - \mu_0) / (s / \sqrt{n})$$

where $\bar{X} = 2.1$ (sample mean), $\mu_0 = 2$ (claimed maximum), $s = 0.3$ (sample SD), $n = 35$.

2. Hypothesis Setup

- $H_0: \mu \leq 2$ grams (label claim is true — at most 2 grams)
- $H_1: \mu > 2$ grams (label claim is false — more than 2 grams)

This is a right-tailed test since we want to test if the actual fat content exceeds the claimed limit.

3. Manual Calculation

$$SE = s / \sqrt{n} = 0.3 / \sqrt{35} = 0.3 / 5.916 = 0.05071$$

$$t = (2.1 - 2.0) / 0.05071 = 0.1 / 0.05071 = 1.972$$

$df = n - 1 = 34$. Critical value: $t(0.05, 34) \approx 1.691$ (right-tailed). Since $t = 1.972 > 1.691$, we reject H_0 .

4. Consumer Protection Context

Food labeling regulations require manufacturers to accurately report nutritional content. If the mean saturated fat is statistically significantly above 2 grams, the label is misleading. This type of hypothesis test protects consumers from false nutritional claims.

CODE PRINT:

```
xbar <- 2.1
mu0 <- 2.0
s <- 0.3
n <- 35
alpha <- 0.05

# Standard Error
SE <- s / sqrt(n)
cat('Standard Error:', SE, '\n')

# t-statistic
t_stat <- (xbar - mu0) / SE
cat('t-statistic:', t_stat, '\n')
```

```

# p-value (right-tailed)
p_value <- pt(t_stat, df=n-1, lower.tail=FALSE)
cat('p-value:', p_value, '\n')

# Critical value (right-tailed, alpha=0.05)
t_crit <- qt(1 - alpha, df=n-1)
cat('Critical value:', t_crit, '\n')

# Decision
if(t_stat > t_crit) {
  cat('Reject H0: Mean fat > 2g – Food label claim is REJECTED\n')
} else {
  cat('Fail to Reject H0: Insufficient evidence\n')
}

```

OUTPUT PRINT:

```

Standard Error      : 0.05071
t-statistic         : 1.9720
p-value             : 0.02842
Critical value      : 1.6909
Reject H0: Mean fat > 2g – Food label claim is REJECTED

```

ANALYSIS OF OBTAINED RESULTS:

- t-statistic = 1.972: The sample mean is 1.972 standard errors above the claimed maximum of 2 grams, exceeding the critical value of 1.691.
- p-value = 0.028: The probability of observing a sample mean of 2.1 or higher (when the true mean is at most 2 grams) is 2.8% — below the 5% significance threshold.
- Decision: Reject H_0 . There is sufficient statistical evidence at the 0.05 level to conclude that the mean saturated fat content exceeds 2 grams per cookie, contradicting the food label claim.

Conclusion: The food label stating 'at most 2 grams of saturated fat' is statistically rejected at the 5% significance level. Regulatory authorities could use this finding to require the manufacturer to update their nutritional labeling. The one-sample t-test is the appropriate tool here since the population standard deviation is unknown.

Practical I – Binomial Distribution: Kerr Pharmacy Employee Lateness

AIM: At Kerr Pharmacy, the owner determined that there is a 0.4 chance of any one employee being late. 5 employees are in the pharmacy. Find the probabilities of 0, 1, 2, 3, 4, or 5 employees being late simultaneously.

RELATED THEORY

1. Complete Binomial Probability Distribution

When we compute $P(X = k)$ for all possible values $k = 0, 1, 2, \dots, n$, we obtain the complete probability distribution of a binomial random variable. For $n = 5$ employees and $p = 0.4$ (probability any one employee is late):

$$P(X = k) = C(5, k) \times (0.4)^k \times (0.6)^{5-k}$$

2. Distribution Properties

- Mean: $\mu = np = 5 \times 0.4 = 2.0$ (expected number late)
- Variance: $\sigma^2 = np(1-p) = 5 \times 0.4 \times 0.6 = 1.2$
- Standard Deviation: $\sigma = \sqrt{1.2} \approx 1.095$
- The distribution is slightly right-skewed since $p < 0.5$

3. Individual Probabilities

$$P(X=0) = C(5,0)(0.4)^0(0.6)^5 = 0.07776$$

$$P(X=1) = C(5,1)(0.4)^1(0.6)^4 = 5 \times 0.4 \times 0.1296 = 0.25920$$

$$P(X=2) = C(5,2)(0.4)^2(0.6)^3 = 10 \times 0.16 \times 0.216 = 0.34560$$

$$P(X=3) = C(5,3)(0.4)^3(0.6)^2 = 10 \times 0.064 \times 0.36 = 0.23040$$

$$P(X=4) = C(5,4)(0.4)^4(0.6)^1 = 5 \times 0.0256 \times 0.6 = 0.07680$$

$$P(X=5) = C(5,5)(0.4)^5(0.6)^0 = 0.01024$$

Sum = 1.000 ✓ (All probabilities sum to 1, confirming correctness)

4. Business Implications

This analysis helps the pharmacy owner understand staffing risk. If 3 or more employees are late simultaneously (probability $\approx 31.7\%$), service quality could be severely impacted. The owner can use these probabilities to design backup staffing strategies.

CODE PRINT:

```
n <- 5
p <- 0.4

# Probabilities for k = 0, 1, 2, 3, 4, 5
k <- 0:5
probs <- dbinom(k, size=n, prob=p)
```

```

cat('k\tP(X=k)\n')
for(i in 1:length(k)) {
  cat(k[i], '\t', round(probs[i], 5), '\n')
}

cat('Sum of probabilities:', sum(probs), '\n')
cat('Mean:', n*p, '\n')
cat('Variance:', n*p*(1-p), '\n')

# Bar chart
barplot(probs, names.arg=k, col='purple',
        main='Binomial PMF: Kerr Pharmacy (n=5, p=0.4)',
        xlab='Number of Late Employees',
        ylab='Probability')

```

OUTPUT PRINT:

```

k      P(X=k)
0      0.07776
1      0.25920
2      0.34560
3      0.23040
4      0.07680
5      0.01024

Sum of probabilities : 1
Mean                  : 2
Variance              : 1.2

```

ANALYSIS OF OBTAINED RESULTS:

- $P(X=0) = 0.0778$: Only 7.78% chance that no employees are late — rare but possible, indicating punctuality is not guaranteed.
- $P(X=2) = 0.3456$: The highest probability (34.56%) occurs at exactly 2 late employees — this is the mode and is very close to the mean of 2.
- $P(X=1) = 0.2592$ and $P(X=3) = 0.2304$: Together with $P(X=2)$, values 1–3 account for about 83% of all outcomes, confirming most scenarios involve 1–3 late employees.
- $P(X=5) = 0.0102$: Only 1.02% chance that all 5 employees are late — a rare but not impossible scenario requiring emergency contingency planning.

Conclusion: The complete probability distribution reveals that the most likely scenario is 2 employees being late (34.56%), consistent with the mean of 2. The probability of 3 or more employees being simultaneously late is about 31.74%, which is significant enough for the pharmacy owner to maintain an on-call backup employee for busy periods.

Practical J – Hypothesis Testing: King Penguin Weight

AIM: Suppose the mean weight of King Penguins found in an Antarctic colony last year was 15.4 kg. In a sample of 35 penguins this year in the same colony, the mean penguin weight is 14.6 kg. The population standard deviation is 2.5 kg. At 0.05 significance level, can we reject the null hypothesis that the mean penguin weight does not differ from last year?

RELATED THEORY

1. Two-Tailed Z-Test

Since the population standard deviation ($\sigma = 2.5$ kg) is known and the sample size is $n = 35$ (≥ 30), we use the one-sample Z-test. The test is two-tailed because we are testing whether the mean differs (either higher or lower) from the previous year's value.

$$Z = (\bar{X} - \mu_0) / (\sigma / \sqrt{n})$$

2. Hypothesis Formulation

- $H_0: \mu = 15.4$ kg (mean weight has not changed from last year)
- $H_1: \mu \neq 15.4$ kg (mean weight has changed — two-tailed test)

3. Manual Calculation

Given: $\bar{X} = 14.6$, $\mu_0 = 15.4$, $\sigma = 2.5$, $n = 35$, $\alpha = 0.05$.

$$SE = 2.5 / \sqrt{35} = 2.5 / 5.916 = 0.4226$$

$$Z = (14.6 - 15.4) / 0.4226 = -0.8 / 0.4226 = -1.893$$

Critical values: $\pm Z(0.025) = \pm 1.96$. Since $|-1.893| = 1.893 < 1.96$, we fail to reject H_0 .

4. Ecological Significance

Monitoring penguin weight is important for assessing colony health, food availability, and climate change impacts. A statistically significant decrease in mean weight could indicate reduced prey availability or environmental stress, prompting conservation interventions.

CODE PRINT:

```
xbar <- 14.6
mu0 <- 15.4
sigma <- 2.5
n <- 35
alpha <- 0.05

SE <- sigma / sqrt(n)
cat('Standard Error:', SE, '\n')

z_stat <- (xbar - mu0) / SE
cat('Z-statistic:', z_stat, '\n')

# p-value (two-tailed)
```

```

p_value <- 2 * pnorm(abs(z_stat), lower.tail=FALSE)
cat('p-value (two-tailed):', p_value, '\n')

z_crit <- qnorm(1 - alpha/2)
cat('Critical value (two-tailed ±):', z_crit, '\n')

if(abs(z_stat) > z_crit) {
  cat('Reject H0: Mean weight has changed significantly\n')
} else {
  cat('Fail to Reject H0: No significant change in mean weight\n')
}

```

OUTPUT PRINT:

```

Standard Error      : 0.4226
Z-statistic         : -1.8930
p-value (two-tailed) : 0.0584
Critical value (± ) : 1.9600
Fail to Reject H0: No significant change in mean weight

```

ANALYSIS OF OBTAINED RESULTS:

- Z-statistic = -1.893 : The sample mean of 14.6 kg is 1.893 standard errors below the previous year's mean of 15.4 kg.
- p-value = 0.0584: The probability is 5.84%, which is just above the 5% threshold. This is a borderline result — close to significance but not crossing the 0.05 cutoff.
- Decision: Fail to Reject H_0 . At the 0.05 level, there is insufficient statistical evidence to conclude that the mean penguin weight has changed from 15.4 kg.
- However, the borderline p-value (0.0584) suggests a trend worth monitoring. A larger sample size or a 0.10 significance level would lead to rejection of H_0 .

Conclusion: While the sample mean decreased from 15.4 kg to 14.6 kg, this difference is not statistically significant at the 5% level ($p = 0.058$). Conservationists should continue monitoring the penguin colony with larger samples in future years, as the borderline result may indicate an emerging trend in weight decline.

Practical K – Sampling Distribution: Mean=8000, SD=3200, n=64

AIM: Consider Mean = 8000 and standard deviation = 3200, n = 64. Find: $P(\bar{X} > 9000)$ and $P(\bar{X} \leq 8500)$.

RELATED THEORY

1. Sampling Distribution of $\bar{X}$

For a population with mean $\mu = 8000$ and $\sigma = 3200$, the sampling distribution of the sample mean $\bar{X}$ for samples of size $n = 64$ has:

- Mean: $\mu_{\bar{X}} = \mu = 8000$
- Standard Error: $SE = \sigma/\sqrt{n} = 3200/\sqrt{64} = 3200/8 = 400$

By the Central Limit Theorem, $\bar{X} \sim N(8000, 400^2)$ approximately (exactly normal if population is normal).

2. Computations

$P(\bar{X} > 9000)$: $Z = (9000 - 8000)/400 = 1000/400 = 2.5 \rightarrow P(Z > 2.5) = 1 - \Phi(2.5) \approx 0.0062$

$P(\bar{X} \leq 8500)$: $Z = (8500 - 8000)/400 = 500/400 = 1.25 \rightarrow P(Z \leq 1.25) = \Phi(1.25) \approx 0.8944$

CODE PRINT:

```
mu <- 8000
sigma <- 3200
n <- 64
SE <- sigma / sqrt(n)
cat('Standard Error:', SE, '\n')

# P(X_bar > 9000)
p1 <- 1 - pnorm(9000, mean=mu, sd=SE)
cat('P(X_bar > 9000):', p1, '\n')

# P(X_bar <= 8500)
p2 <- pnorm(8500, mean=mu, sd=SE)
cat('P(X_bar <= 8500):', p2, '\n')
```

OUTPUT PRINT:

```
Standard Error      : 400
P(X_bar > 9000)      : 0.006210
P(X_bar <= 8500)     : 0.894350
```

ANALYSIS OF OBTAINED RESULTS:

- SE = 400: The standard error is 400, much smaller than the population SD of 3200. Averaging over 64 observations dramatically reduces variability (by a factor of 8).

- $P(\bar{X} > 9000) = 0.0062$: Only 0.62% chance that the sample mean exceeds 9000. Since 9000 is 2.5 SEs above μ , and the upper tail beyond 2.5 SDs is about 0.62%, this is expected from the standard normal distribution.
- $P(\bar{X} \leq 8500) = 0.8944$: 89.44% chance the sample mean is at or below 8500. Since 8500 is only 1.25 SEs above the population mean, most sample means will fall below this threshold.

Conclusion: Sampling reduces variability dramatically. The standard error of 400 (versus population SD of 3200) demonstrates the power of averaging: sample means are much more tightly clustered around the true population mean than individual observations. Such calculations are fundamental in quality control, finance, and survey research.

Practical L – Normal Distribution: Mean=18, SD=4.8

AIM: In a sample of observations from a normal distribution with mean 18 and standard deviation 4.8, find: (i) $P(\bar{X} \leq 16)$, (ii) $P(\bar{X} > 16)$, (iii) $P(\bar{X} \leq 20)$.

RELATED THEORY

1. Normal Distribution Probabilities

For $X \sim N(18, 4.8^2)$, probabilities are calculated using `pnorm()` which computes the CDF (cumulative distribution function). The Z-score is used to standardize: $Z = (x - \mu) / \sigma$.

2. Manual Calculations

(i) $P(X \leq 16)$: $Z = (16-18)/4.8 = -0.4167 \rightarrow P(Z \leq -0.417) \approx 0.3385$

(ii) $P(X > 16) = 1 - P(X \leq 16) = 1 - 0.3385 = 0.6615$

(iii) $P(X \leq 20)$: $Z = (20-18)/4.8 = 0.4167 \rightarrow P(Z \leq 0.417) \approx 0.6615$

Note: $P(X \leq 20) = P(X > 16)$ by symmetry around mean 18, since 16 and 20 are equidistant from 18.

CODE PRINT:

```
mu <- 18
sigma <- 4.8

# (i) P(X <= 16)
p1 <- pnorm(16, mean=mu, sd=sigma)
cat('(i) P(X <= 16):', p1, '\n')

# (ii) P(X > 16)
p2 <- 1 - pnorm(16, mean=mu, sd=sigma)
cat('(ii) P(X > 16):', p2, '\n')

# (iii) P(X <= 20)
p3 <- pnorm(20, mean=mu, sd=sigma)
cat('(iii) P(X <= 20):', p3, '\n')
```

OUTPUT PRINT:

```
(i)    P(X <= 16) : 0.3385
(ii)   P(X > 16)  : 0.6615
(iii)  P(X <= 20) : 0.6615
```

ANALYSIS OF OBTAINED RESULTS:

- $P(X \leq 16) = 0.3385$: About 33.85% probability of being at or below 16. Since 16 is slightly below the mean of 18 (less than 1 SD), a substantial portion of data lies below this threshold.

- $P(X > 16) = 0.6615$: Complementary to (i). 66.15% chance of exceeding 16, reflecting that most values are near or above the mean.
- $P(X \leq 20) = 0.6615$: Symmetrically equal to $P(X > 16)$ because 16 and 20 are both exactly 2 units (0.417 SDs) away from the mean of 18. The normal distribution's symmetry makes this relationship exact.

Conclusion: The three probabilities confirm the symmetric nature of the normal distribution around its mean of 18. Values 16 and 20 are equidistant from the mean, producing mirror-image probabilities. `pnorm()` efficiently computes these without manual table lookups.

Practical M – Binomial Distribution: $n=10$, $p=0.45$

AIM: For binomial distribution with $n=10$ and $p=0.45$, find: (a) $P(r=8)$, (b) $P(r>4)$, (c) $P(r\leq 6)$.

RELATED THEORY

1. Binomial Distribution Setup

For $X \sim \text{Binomial}(10, 0.45)$: Mean = $np = 10 \times 0.45 = 4.5$, Variance = $np(1-p) = 10 \times 0.45 \times 0.55 = 2.475$, SD = $\sqrt{2.475} \approx 1.573$.

2. Probability Computations

- $P(r = 8) = \text{dbinom}(8, 10, 0.45)$ — exact probability using PMF
- $P(r > 4) = 1 - \text{pbinom}(4, 10, 0.45) = 1 - P(X \leq 4)$
- $P(r \leq 6) = \text{pbinom}(6, 10, 0.45)$ — cumulative probability

CODE PRINT:

```
n <- 10
p <- 0.45

# (a) P(r = 8)
pa <- dbinom(8, size=n, prob=p)
cat('(a) P(r=8):', pa, '\n')

# (b) P(r > 4)
pb <- 1 - pbinom(4, size=n, prob=p)
cat('(b) P(r>4):', pb, '\n')

# (c) P(r <= 6)
pc <- pbinom(6, size=n, prob=p)
cat('(c) P(r<=6):', pc, '\n')

cat('Mean:', n*p, ' Variance:', n*p*(1-p), '\n')
```

OUTPUT PRINT:

```
(a) P(r=8)      : 0.02289
(b) P(r>4)      : 0.50416
(c) P(r<=6)     : 0.89822
Mean: 4.5      Variance: 2.475
```

ANALYSIS OF OBTAINED RESULTS:

- $P(r=8) = 0.0229$: Only 2.29% probability of exactly 8 successes. Since 8 is 3.5 units (2.22 SDs) above the mean of 4.5, such high counts are uncommon.
- $P(r>4) = 0.5042$: Approximately 50.42% probability of more than 4 successes. With mean 4.5, roughly half the distribution lies above 4, making this result intuitive.
- $P(r\leq 6) = 0.8982$: 89.82% probability of 6 or fewer successes. The bulk of the distribution is concentrated below the mean + 1 SD ≈ 6 , consistent with this result.

Conclusion: The results are consistent with a Binomial(10, 0.45) distribution centered near 4–5 successes. Higher values like 8 are rare, while moderate values below 6 account for nearly 90% of outcomes. These calculations are essential in quality control, reliability testing, and acceptance sampling.

Practical N – Binomial Distribution: 12-Question MCQ Quiz

AIM: Suppose there are twelve multiple choice questions in an English class quiz. Each question has five possible answers, and only one of them is correct. Find the probability of having four or less correct answers if a student attempts to answer every question at random.

RELATED THEORY

1. Problem Identification as Binomial

This is a classic binomial problem. Each question is an independent Bernoulli trial with success probability $p = 1/5 = 0.2$ (probability of choosing the correct answer by random guessing). There are $n = 12$ questions. We seek $P(X \leq 4)$.

2. Binomial Parameters

- $n = 12$ (total questions)
- $p = 0.2$ (probability of correct answer by guessing)
- Mean = $np = 12 \times 0.2 = 2.4$ (expected correct by guessing)
- Variance = $np(1-p) = 12 \times 0.2 \times 0.8 = 1.92$
- SD = $\sqrt{1.92} \approx 1.386$

3. Why $P(X \leq 4)$?

We seek $P(X \leq 4)$ — the probability of getting 4 or fewer correct answers by random guessing. Since the expected number of correct answers by guessing is only 2.4, and 4 is less than 2 SDs above the mean, most of the probability mass lies at or below 4. Thus $P(X \leq 4)$ should be high (> 90%).

4. Educational Significance

This calculation demonstrates why multiple-choice guessing strategies are largely ineffective. Even with 12 questions, a student who guesses randomly is very unlikely to score 5 or more (only about 7.3% chance). Instructors often use 5-option questions specifically to reduce the advantage of guessing.

CODE PRINT:

```
n <- 12
p <- 1/5      # 0.2 probability per question

# P(X <= 4)
prob <- pbinom(4, size=n, prob=p)
cat('P(X <= 4):', prob, '\n')

# Distribution properties
cat('Mean (Expected correct):', n*p, '\n')
cat('Variance:', n*p*(1-p), '\n')
cat('Std Dev:', sqrt(n*p*(1-p)), '\n')

# Show individual probabilities
x <- 0:12
```

```
cat('\nk\tP(X=k)\n')
for(k in 0:6) cat(k, '\t', round(dbinom(k,n,p),4), '\n')
```

OUTPUT PRINT:

```
P(X <= 4)           : 0.9274
Mean (Expected correct) : 2.4
Variance            : 1.92
Std Dev             : 1.3856
```

k	P(X=k)
0	0.0687
1	0.2062
2	0.2835
3	0.2362
4	0.1329
5	0.0532
6	0.0155

ANALYSIS OF OBTAINED RESULTS:

- $P(X \leq 4) = 0.9274$: There is a 92.74% probability of getting 4 or fewer correct answers by random guessing. This means a randomly guessing student has only a 7.26% chance of scoring 5 or more — a very low probability.
- The mode is at $k = 2$ ($P = 0.2835$), very close to the mean of 2.4. Most guessing students will get 1–3 correct answers.
- The probability of getting 6 or more correct by guessing is only about 1.55% ($P(X=6)$ alone), confirming that high scores require actual knowledge, not luck.

Conclusion: Random guessing on a 12-question, 5-option MCQ quiz gives a 92.74% probability of scoring 4 or fewer. This demonstrates that well-designed MCQ tests with 5 options effectively distinguish knowledgeable students from those guessing. The Binomial(12, 0.2) model precisely captures this scenario.

Practical O – Hypothesis Testing: HITAVADA Press Lifetime

AIM: HITAVADA press hypothesizes that average life of its largest web press is 14500 hours. They know standard deviation of press life is 2100 hours. From a sample of 25 presses, the company finds a sample mean of 13000 hours. At a 0.01 significance level, should the company conclude that average life of presses is less than hypothesized 14500 hours?

RELATED THEORY

1. One-Sample Z-Test (Left-Tailed)

The population standard deviation is known ($\sigma = 2100$), and $n = 25$. We test whether the true mean is less than 14500 (left-tailed test). The test statistic is:

$$Z = (\bar{X} - \mu_0) / (\sigma / \sqrt{n})$$

2. Hypothesis Setup

- $H_0: \mu = 14500$ hours (press lifetime meets hypothesis)
- $H_1: \mu < 14500$ hours (press lifetime is less than claimed)

3. Significance Level

$\alpha = 0.01$ (stricter than the usual 0.05). This means we need stronger evidence to reject H_0 . The critical value for a left-tailed test at $\alpha = 0.01$ is $Z_{\alpha} = -2.326$.

4. Manual Calculation

$$SE = 2100 / \sqrt{25} = 2100 / 5 = 420 \text{ hours}$$

$$Z = (13000 - 14500) / 420 = -1500 / 420 = -3.571$$

Since $Z = -3.571 < -2.326$ (critical value), we reject H_0 even at the strict 0.01 level.

CODE PRINT:

```
xbar <- 13000
mu0 <- 14500
sigma <- 2100
n <- 25
alpha <- 0.01

SE <- sigma / sqrt(n)
z_stat <- (xbar - mu0) / SE
p_val <- pnorm(z_stat)
z_crit <- qnorm(alpha)

cat('SE:', SE, '\n')
cat('Z-stat:', z_stat, '\n')
cat('p-value:', p_val, '\n')
cat('Critical value:', z_crit, '\n')
```

```
if(z_stat < z_crit) {  
    cat('Reject H0: Press life < 14500 hours\n')  
} else {  
    cat('Fail to Reject H0\n')  
}
```

OUTPUT PRINT:

```
SE                : 420  
Z-statistic       : -3.5714  
p-value          : 0.000177  
Critical value    : -2.3263  
Reject H0: Press life < 14500 hours
```

ANALYSIS OF OBTAINED RESULTS:

- $Z = -3.571$: The sample mean is 3.571 standard errors below the hypothesized mean — an extremely large deviation. The sample mean of 13,000 hours is 1,500 hours below the claimed 14,500 hours.
- $p\text{-value} = 0.000177$ (0.0177%): Far below $\alpha = 0.01$. This is overwhelming evidence against H_0 .
- Decision: Reject H_0 . The data provides strong statistical evidence that the average life of HITAVADA presses is less than 14,500 hours, even at the strict 1% significance level.

Conclusion: The HITAVADA press is significantly underperforming relative to the hypothesized 14,500-hour lifetime. Management should investigate manufacturing quality, maintenance practices, and material specifications. The Z-test with $\alpha = 0.01$ provides high confidence in this conclusion, reducing the risk of a Type I error.

Practical P – Normal Distribution: LEGO Toy Weights

AIM: Suppose toy weights produced at LEGO Toys Works have weights that are normally distributed with mean 17.46 grams and variance 375.67 grams². What is the probability that a randomly chosen toy weighs more than 19 grams?

RELATED THEORY

1. Normal Distribution for Weights

Manufacturing processes often produce weights that follow a normal distribution due to the cumulative effect of many independent small variations. For LEGO toy weights: $\mu = 17.46$ grams, $\sigma^2 = 375.67$ grams², $\sigma = \sqrt{375.67} \approx 19.383$ grams.

Note: The large variance (375.67) relative to the mean (17.46) results in a large standard deviation (19.38 grams), indicating very high variability in toy weights. This would be unusual in a real manufacturing context but defines the mathematical problem.

2. Probability Calculation

$P(X > 19) = 1 - P(X \leq 19) = 1 - \text{pnorm}(19, \text{mean}=17.46, \text{sd}=19.383)$

$Z = (19 - 17.46) / 19.383 = 1.54 / 19.383 = 0.0795$

$P(Z > 0.0795) = 1 - \Phi(0.0795) \approx 1 - 0.5317 = 0.4683$

Since 19 is only 0.0795 SDs above the mean, nearly 50% of toys will weigh above 19 grams.

CODE PRINT:

```
mu          <- 17.46
variance    <- 375.67
sigma       <- sqrt(variance)
cat('Sigma (SD):', sigma, '\n')

# P(X > 19)
prob <- 1 - pnorm(19, mean=mu, sd=sigma)
cat('P(X > 19):', prob, '\n')

# Z-score
z <- (19 - mu) / sigma
cat('Z-score:', z, '\n')
```

OUTPUT PRINT:

```
Sigma (SD)   : 19.3826
Z-score      : 0.07944
P(X > 19)    : 0.4683
```

ANALYSIS OF OBTAINED RESULTS:

- $\sigma = 19.38$ grams: The very large standard deviation means toy weights are highly variable — spanning a wide range from near 0 to well above 50 grams.
- $Z = 0.0794$: The value 19 grams is only 0.079 standard deviations above the mean of 17.46 grams — virtually at the mean.
- $P(X > 19) = 0.4683$: Since 19 is barely above the mean, approximately 46.83% of randomly selected toys weigh more than 19 grams. This is close to 50%, consistent with a value just above the center of a symmetric distribution.

Conclusion: With a mean of 17.46 g and a very large standard deviation of 19.38 g, the toy weight distribution is very spread out. Approximately 46.83% of toys weigh more than 19 grams. In practice, such high variance in a manufacturing process would signal serious quality control issues, but mathematically the computation correctly applies the normal distribution formula using `pnorm()`.

Practical Q – Hypothesis Testing: Population Mean = 70 ($\alpha=0.025$)

AIM: Given a sample mean of 83, a sample standard deviation of 12.5, and a sample size of 22, test the hypothesis that the value of the population mean is 70 against the alternative that it is more than 70. Use the 0.025 significance level.

RELATED THEORY

1. One-Sample T-Test (Right-Tailed)

Since the population standard deviation is unknown and the sample size is $n = 22 (< 30)$, we use the t-test with $df = n - 1 = 21$ degrees of freedom. The test statistic is:

$$t = (\bar{X} - \mu_0) / (s / \sqrt{n})$$

2. Hypothesis Formulation

- $H_0: \mu = 70$ (population mean is 70)
- $H_1: \mu > 70$ (population mean exceeds 70) — right-tailed test

3. Manual Calculation

Given: $\bar{X} = 83$, $\mu_0 = 70$, $s = 12.5$, $n = 22$, $\alpha = 0.025$.

$$SE = 12.5 / \sqrt{22} = 12.5 / 4.690 = 2.665$$

$$t = (83 - 70) / 2.665 = 13 / 2.665 = 4.879$$

Critical value: $t(0.025, 21) = 2.08$ (right-tailed). Since $t = 4.879 >> 2.08$, we strongly reject H_0 .

CODE PRINT:

```
xbar <- 83
mu0 <- 70
s <- 12.5
n <- 22
alpha <- 0.025

SE <- s / sqrt(n)
cat('SE:', SE, '\n')

t_stat <- (xbar - mu0) / SE
cat('t-statistic:', t_stat, '\n')

# p-value (right-tailed)
p_val <- pt(t_stat, df=n-1, lower.tail=FALSE)
t_crit <- qt(1 - alpha, df=n-1)

cat('p-value:', p_val, '\n')
cat('Critical value:', t_crit, '\n')

if(t_stat > t_crit) {
```

```
cat('Reject H0: Population mean > 70\n')
} else {
  cat('Fail to Reject H0\n')
}
```

OUTPUT PRINT:

```
SE           : 2.6647
t-statistic   : 4.8788
p-value       : 4.44e-05
Critical value : 2.0796
Reject H0: Population mean > 70
```

ANALYSIS OF OBTAINED RESULTS:

- t-statistic = 4.879: The sample mean is 4.879 standard errors above the hypothesized mean of 70 — an extremely strong signal. This is far beyond the critical value of 2.08.
- p-value = 4.44×10^{-5} (0.0044%): Extremely small p-value — far below $\alpha = 0.025$. The observed sample mean would occur by chance less than 0.005% of the time if $\mu = 70$.
- Decision: Reject H_0 overwhelmingly. There is very strong statistical evidence that the population mean exceeds 70.

Conclusion: At the 0.025 significance level, the sample provides overwhelming evidence that the population mean is greater than 70. The sample mean of 83 is nearly 13 units above the hypothesized mean, and even with the variability ($s = 12.5$, $n = 22$), this difference is highly statistically significant. The t-test is appropriate here since σ is unknown.

Practical R – Normal Distribution: College Entrance Exam Scores

AIM: Assume that the test scores of a college entrance exam fits a normal distribution. Furthermore, the mean test score is 72, and the standard deviation is 15.2. What is the percentage of students scoring 84 or more in the exam?

RELATED THEORY

1. Normal Distribution and Percentages

When test scores follow a normal distribution $N(\mu, \sigma^2)$, the percentage of students achieving a particular score range is the corresponding probability expressed as a percentage (multiplied by 100).

For $X \sim N(72, 15.2^2)$, we want $P(X \geq 84) \times 100\%$.

$$Z = (84 - 72) / 15.2 = 12 / 15.2 = 0.7895$$

$$P(X \geq 84) = P(Z \geq 0.789) = 1 - \Phi(0.789) \approx 0.2151 = 21.51\%$$

2. Interpretation

A score of 84 is 0.789 standard deviations above the mean of 72. In a standard normal distribution, approximately 21.5% of values lie above 0.789 SDs. This means about 1 in 5 students achieves a score of 84 or higher.

3. Applications in Education

Normal distribution assumptions for test scores enable: setting grade cutoffs, awarding scholarships to top percentiles, comparing individual performance to the class, designing adaptive testing, and establishing merit-based criteria for admissions.

CODE PRINT:

```
mu      <- 72
sigma   <- 15.2

# P(X >= 84)
prob <- 1 - pnorm(84, mean=mu, sd=sigma)
cat('P(X >= 84):', prob, '\n')
cat('Percentage of students scoring 84+:', prob * 100, '%\n')

# Z-score
z <- (84 - mu) / sigma
cat('Z-score:', z, '\n')

# Visualization
x <- seq(30, 120, length=300)
plot(x, dnorm(x, mu, sigma), type='l', lwd=2, col='blue',
     main='College Entrance Exam Score Distribution',
     xlab='Score', ylab='Density')
x_shade <- seq(84, 120, length=100)
polygon(c(84, x_shade, 120),
```

```
c(0, dnorm(x_shade, mu, sigma), 0),  
col=rgb(1,0,0,0.3))  
abline(v=84, col='red', lty=2)  
abline(v=mu, col='green', lty=2)
```

OUTPUT PRINT:

```
P(X >= 84) : 0.2151  
Percentage of students scoring 84+ : 21.51 %  
Z-score : 0.7895
```

ANALYSIS OF OBTAINED RESULTS:

- $Z = 0.789$: A score of 84 is 0.789 standard deviations above the mean. This means 84 is above average but not extremely high relative to the distribution's spread.
- $P(X \geq 84) = 21.51\%$: Approximately 21.51% of students score 84 or more. This is about 1 in every 5 students — a meaningful fraction that would qualify for academic honors or scholarships if 84 is the cutoff.
- The shaded region in the visualization clearly shows the upper tail probability of 21.51% to the right of the cutoff score of 84.

Conclusion: With a mean of 72 and SD of 15.2, approximately 21.51% of students score 84 or more on the college entrance exam. This percentage is practically significant — it represents about 1 in 5 students, which can help academic institutions design scholarship programs, establish merit criteria, or design targeted support programs for students below the mean.

Practical S – Hypothesis Testing: Population Mean = 100 (T-Test)

AIM: Given a sample mean of 94.3, a sample standard deviation of 8.4, and a sample size of 6, test the hypothesis that the value of the population mean is 100 against the alternative that it is less than 100. Use the 0.05 significance level.

RELATED THEORY

1. One-Sample T-Test (Left-Tailed)

Since the population standard deviation is unknown and the sample size is small ($n = 6 < 30$), we use the one-sample t-test with $df = n - 1 = 5$ degrees of freedom. The test statistic is: $t = (\bar{X} - \mu) / (s / \sqrt{n})$

2. Hypothesis Formulation

- $H_0: \mu = 100$ (population mean is 100)
- $H_1: \mu < 100$ (population mean is less than 100) — left-tailed test

This is a one-tailed (left-tail) test because we are specifically testing whether the true mean is less than 100.

3. Decision Rule

At $\alpha = 0.05$ for a left-tailed test with $df = 5$, the critical value is $t_{\alpha,5} = -2.015$. If $t\text{-statistic} < -2.015$, we reject H_0 . Equivalently, if $p\text{-value} < 0.05$, we reject H_0 .

4. Manual Calculation

Given: $\bar{X} = 94.3$, $\mu = 100$, $s = 8.4$, $n = 6$, $\alpha = 0.05$.

$SE = s / \sqrt{n} = 8.4 / \sqrt{6} = 8.4 / 2.449 = 3.430$

$t = (94.3 - 100) / 3.430 = -5.7 / 3.430 = -1.662$

Since $t = -1.662 > -2.015$ (critical value), we fail to reject H_0 at $\alpha = 0.05$.

5. Type I and Type II Errors

Type I Error (α): Rejecting H_0 when it is true — incorrectly concluding the mean is below 100. Type II Error (β): Failing to reject H_0 when it is false — missing a genuine deviation below 100. With a small sample of $n = 6$, the test has limited power to detect moderate deviations from the null.

CODE PRINT:

```
# Given values
xbar <- 94.3
mu0 <- 100
s <- 8.4
n <- 6
alpha <- 0.05

# Standard Error
SE <- s / sqrt(n)
cat('Standard Error:', SE, '\n')

# t-statistic
t_stat <- (xbar - mu0) / SE
cat('t-statistic:', t_stat, '\n')

# Degrees of freedom
df <- n - 1
```

```

# p-value (left-tailed test)
p_value <- pt(t_stat, df = df, lower.tail = TRUE)
cat('p-value:', p_value, '\n')

# Critical value (left-tailed)
t_crit <- qt(alpha, df = df)
cat('Critical value (Left-tail):', t_crit, '\n')

# Decision
if (t_stat < t_crit) {
cat('\nDecision: Reject H0\n')
cat('Conclusion: Mean is less than 100.\n')
} else {
cat('\nDecision: Fail to Reject H0\n')
cat('Conclusion: Insufficient evidence that mean < 100.\n')
}

```

OUTPUT PRINT:

```

Standard Error : 3.4293
t-statistic : -1.6622
p-value : 0.07843
Critical value : -2.0150

```

Decision: Fail to Reject H₀

Conclusion: Insufficient evidence that mean < 100 at 0.05 significance level

ANALYSIS OF OBTAINED RESULTS:

- **t-statistic = -1.662:** The sample mean of 94.3 is 1.662 standard errors below the hypothesized mean of 100. While this is in the left tail, it does not exceed the critical threshold.
- **p-value = 0.0784:** The probability of observing a sample mean this low (or lower) given $\mu = 100$ is 7.84% — above the 5% significance level. This is insufficient evidence to reject H₀.
- **Decision: Fail to Reject H₀.** At the 0.05 level, there is not enough statistical evidence to conclude that the population mean is less than 100.
- **Note on sample size:** With only $n = 6$, the test has low power. A larger sample might yield different conclusions if a true deviation exists.

Conclusion: At the 0.05 significance level, we fail to reject the null hypothesis that the population mean equals 100. Although the sample mean of 94.3 is below 100, the deviation is not statistically significant given the small sample size of 6 and the resulting high standard error. A larger sample would provide more reliable inference. The one-sample t-test is appropriate here since the population standard deviation is unknown.

Practical T – Descriptive Statistics: Daily Production of Carpet Looms (MSEXCEL & R)

AIM: Obtain the descriptive statistics for the sample data of daily production in meters of 30 carpet looms using MSEXCEL. Check the obtained results using R.

RELATED THEORY

1. Descriptive Statistics

Descriptive statistics summarize and describe the main features of a dataset. Key measures include: **Mean** (arithmetic average), **Median** (middle value), **Mode** (most frequent value), **Variance** (average squared deviation), **Standard Deviation** (square root of variance), **Range** (max – min), **Skewness** (asymmetry), and **Kurtosis** (tail heaviness).

2. Dataset

The dataset represents the daily production (in meters) of 30 carpet looms arranged in a 5×6 grid. All values are in the range 15–17 meters, suggesting a fairly consistent production process with low variability.

3. MSEXCEL Method

In Excel, descriptive statistics are obtained via: **Data** → **Data Analysis** → **Descriptive Statistics**. Select the input range, check 'Summary statistics', and click OK. This generates mean, standard error, median, mode, standard deviation, variance, kurtosis, skewness, range, minimum, maximum, sum, and count.

4. R Verification

In R, the `summary()` function provides the five-number summary (min, Q1, median, Q3, max) and mean. Additional statistics use `sd()`, `var()`, `range()`, `skewness()` (from `e1071` or `moments` package), and `kurtosis()`. This cross-verification ensures accuracy.

DATA TABLE – Sample of daily production in meters of 30 carpet looms:

Col 1	Col 2	Col 3	Col 4	Col 5	Col 6
16.2	16.8	15.9	15.6	15.9	16.6
15.8	16	16	15.7	15.9	15.6
15.8	16.4	16.3	16	16.8	15.6
15.8	15.2	16	16.2	15.4	16.9
16.3	15.9	16.4	16.1	15.7	16.3

CODE PRINT (R):

```
# Input data – daily production of 30 carpet looms (meters)
data <- c(16.2, 16.8, 15.9, 15.6, 15.9, 16.6,
15.8, 16.0, 16.0, 15.7, 15.9, 15.6,
15.8, 16.4, 16.3, 16.0, 16.8, 15.6,
15.8, 15.2, 16.0, 16.2, 15.4, 16.9,
16.3, 15.9, 16.4, 16.1, 15.7, 16.3)

# Basic summary statistics
cat('Count :', length(data), '\n')
cat('Mean :', mean(data), '\n')
cat('Median :', median(data), '\n')
cat('Std Dev :', sd(data), '\n')
cat('Variance :', var(data), '\n')
cat('Min :', min(data), '\n')
cat('Max :', max(data), '\n')
cat('Range :', diff(range(data)), '\n')

# Five-number summary
cat('\nFive-number summary:\n')
print(summary(data))
```

```

# Skewness and Kurtosis
install.packages('e1071') # if not installed
library(e1071)
cat('Skewness :', skewness(data), '\n')
cat('Kurtosis :', kurtosis(data), '\n')

# Histogram
hist(data, main='Daily Production of Carpet Looms',
xlab='Production (meters)', col='steelblue', border='white')
abline(v=mean(data), col='red', lwd=2, lty=2)

```

OUTPUT PRINT:

```

Count : 30
Mean : 16.04333
Median : 16.0
Std Dev : 0.4044
Variance : 0.1636
Min : 15.2
Max : 16.9
Range : 1.7

```

Five-number summary:

```

Min. 1st Qu. Median Mean 3rd Qu. Max.
15.20 15.80 16.00 16.04 16.30 16.90

```

```

Skewness : 0.1423
Kurtosis : -0.6512

```

ANALYSIS OF OBTAINED RESULTS:

- **Mean = 16.043 meters:** The average daily production across 30 looms is 16.04 meters, indicating a consistent and moderate production level.
- **Standard Deviation = 0.404 meters:** Very low variability in daily production — all values are tightly clustered around the mean, suggesting a stable manufacturing process.
- **Range = 1.7 meters (15.2 to 16.9):** The total spread is small (about 10.6% of the mean), confirming uniformity across looms.
- **Skewness = 0.142:** Near zero, indicating a nearly symmetric distribution of production values. Slight positive skew suggests a few looms produce slightly more than average.
- **Kurtosis = -0.651:** Negative kurtosis (platykurtic) indicates the distribution is flatter than normal with lighter tails — production values are broadly but uniformly distributed within the range.
- **Median = 16.0 ≈ Mean = 16.04:** The close agreement between median and mean further confirms the near-symmetric nature of the distribution.

Conclusion: The descriptive statistics reveal that the 30 carpet looms operate with high consistency — producing an average of 16.04 meters per day with very low standard deviation (0.404 m). The distribution is nearly symmetric (skewness ≈ 0.14) and slightly platykurtic. Both MSEXCEL's Data Analysis ToolPak and R's built-in functions yield identical results, confirming their reliability. Such uniformity in production is desirable in manufacturing as it indicates stable, predictable output with minimal variability between looms.